



Module 11 Day 2: Cryptography

Activity File: GPG

In this activity, you will continue to be a security analyst working for the Hill Valley Police Department.

- Captain Strickland has implemented your suggestion of using asymmetric key encryption across all departments at Hill Valley.
- Captain Strickland believes that the Alphabet Bandit may actually be an insider at the Hill Valley Police Department.
- They would like you to propose a way to catch the insider and communicate this to your partner.

Note: The idea you select isn't super important and is not the focus of this exercise. Rather, the point is to practice using GPG.

Instructions

1. Choose a partner.
2. Write an updated message, using your code name, that describes your idea to catch the insider titled `secret_idea.txt`.
 - For example, an idea might be: *Wiretapping all the detective phones.*
3. Generate a key pair, export your public key with armor and share it with your partner.
4. Import your partner's public key.
5. Encrypt the message with your partner's public key using GPG, save the encrypted file as `[yourname]_secret_idea.txt.enc`.
6. Transmit the encrypted message to your partner using Slack.
 - Note: You will want to use Slack within your virtual machine. To make sure firefox is updated to support Slack, run the following two command line commands:
 - `sudo apt update`
 - `sudo apt upgrade`
7. After you receive your partner's encrypted message, decrypt it using your private key.